RAPIVOLT

On-Demand EV Charging Platform

1. WHAT – What is RapiVolt?

RapiVolt is an on-demand Electric Vehicle (EV) charging platform that connects EV drivers with nearby registered homeowners and professional charging providers.

The platform allows drivers to find available charging facilities, compare prices, and request charging services based on their location. Providers can register their charging facilities, manage availability, set prices, and accept or reject incoming requests.

Initially, RapiVolt will operate as a single application with two different user roles:

- **Driver:** Users who need charging services for their electric vehicles.
- **Charging Provider:** Homeowners and professional charging businesses that offer charging facilities.

The main objective is to create a convenient connection between people who need EV charging and those who can provide it.

2. WHY – Why are we building RapiVolt?

With the increasing adoption of electric vehicles, finding convenient and accessible charging facilities remains a challenge.

EV drivers may face difficulties finding nearby charging stations, especially when their vehicle battery is running low or public charging stations are unavailable or far away.

At the same time, homeowners and private charging providers with suitable charging facilities may have the opportunity to earn additional income by offering charging services.

RapiVolt aims to address these challenges by:

- Making nearby charging facilities easier to find.
- Connecting drivers with available charging providers.
- Providing transparent pricing before booking.
- Reducing dependency on traditional public charging stations.
- Creating earning opportunities for homeowners and charging businesses.
- Supporting the growth of electric mobility.

3. HOW – How will RapiVolt work?

RapiVolt will operate through a location-based service and booking system.

Step 1: User Registration

Users register and log in to the application. They can use the platform as a Driver or Charging Provider. A single account may support both roles.

Step 2: Provider Registration

Charging providers register their facilities by adding their location, charger details, pricing, operating hours and availability.

Step 3: Finding Nearby Providers

Drivers share their current location. The platform identifies nearby available charging providers and displays their distance, charging details, compatibility and prices.

Step 4: Booking Request

Drivers select a suitable provider and send a charging request.

Step 5: Provider Response

The provider receives the request and can accept or reject it according to their availability.

Step 6: Booking Confirmation

Once the provider accepts the request, the booking is confirmed. The driver receives the booking details and relevant location and access instructions.

Step 7: Charging and Payment

The driver receives the charging service. After the charging session, the payment and booking status are recorded.

4. CHARGING EXPERIENCE

RapiVolt will offer two charging service options:

A. Wait & Charge

Drivers can stay nearby while their vehicles are being charged.

During this time, the platform can help them discover nearby:

- Cafes and restaurants.
- Shopping centres and stores.
- Tourist attractions and other places of interest.

Drivers can also receive charging completion notifications and view estimated charging duration and cost.

B. Drop & Charge

Drivers can leave their vehicles with eligible verified charging providers and return after the charging session is completed.

This option will require secure vehicle parking, proper vehicle handover procedures and suitable liability and insurance arrangements.

5. PRIVACY, SAFETY & TRUST

Since RapiVolt connects drivers with residential and commercial charging providers, privacy and security are important considerations.

The platform will include the following measures:

- **Flexible Availability:** Providers can decide their operating hours and manage their availability.
- **Night Charging:** Both residential and commercial providers can offer night-time charging if they choose to enable it.
- **Flexible Access:** Providers can define their charging location and access conditions.
- **User Verification:** Driver and provider identities will be verified to improve trust.
- **Booking Approval:** Providers have complete control over accepting or rejecting requests.
- **Location Privacy:** Exact residential addresses will only be shared after booking confirmation.
- **Emergency Support:** Users will have options to report safety concerns and request assistance.
- **Ratings and Reviews:** Drivers can share feedback about charging providers.

Residential providers will be encouraged to offer separate parking or charging spaces to protect their household privacy.

6. POTENTIAL CHALLENGES & THEIR SOLUTIONS

While developing RapiVolt, several real-world challenges may arise. The platform will aim to address them through the following solutions.

1. Privacy and Security Concerns

Problem: Homeowners may hesitate to allow unknown people to enter their residential property, especially during late-night hours. This can create concerns about family safety and personal privacy.

Solution:

- Mandatory identity verification for users.
- Provider-controlled charging hours and access permissions.
- Preference for separate outdoor charging spaces.
- Sharing residential addresses only after booking confirmation.

- Emergency support and incident reporting.

2. Charging Compatibility Issues

Problem: Different electric vehicles support different charging connectors and power requirements. A provider's charger may not be compatible with a driver's vehicle.

Solution:

- Collect vehicle details during driver registration.
- Collect charger type and power details during provider registration.
- Check vehicle and charger compatibility before booking.
- Display only compatible charging options where compatibility information is verified.

3. Provider Unavailability After Booking

Problem: A provider may accept a booking but become unavailable when the driver reaches the location.

Solution:

- Provider availability confirmation.
- Fixed booking time slots.
- Arrival confirmation.
- Provider cancellation tracking.
- Alternative provider suggestions.

4. Unexpected Pricing and Additional Charges

Problem: Providers may demand additional charges after the booking, creating disputes with drivers.

Solution:

- Display estimated total cost before booking.
- Clearly define per-unit or session-based pricing.
- Lock the agreed pricing for the booking.
- Require driver approval for any additional charges.
- Maintain digital payment records and receipts.

5. Electricity Consumption Disputes

Problem: Homeowners may worry about increased electricity bills, while drivers may question the actual electricity consumed during charging.

Solution:

- Use compatible energy meters or smart charging equipment where available.
- Record actual electricity consumption in kWh.

- Maintain charging session start and end readings.
- Provide transparent charging cost calculations.

6. Booking Cancellation and No-Shows

Problem: Drivers may not arrive after booking, or providers may cancel at the last moment.

Solution:

- Define a clear cancellation policy.
- Send booking reminders.
- Maintain cancellation records.
- Introduce fair cancellation charges where applicable.
- Apply restrictions to repeated misuse of the booking system.

7. Vehicle Theft or Damage During Drop & Charge

Problem: Drivers may hesitate to leave their vehicles with unknown providers due to the risk of theft or damage.

Solution:

- Allow Drop & Charge only through eligible verified providers.
- Record vehicle condition before handover.
- Maintain digital handover and collection confirmation.
- Require suitable secure parking arrangements.
- Establish clear liability and insurance requirements.

8. Unexpected Charging Delays

Problem: Actual charging time may be longer than the estimated duration due to battery condition, charger power or other factors.

Solution:

- Calculate estimated charging duration using vehicle and charger information.
- Update estimated completion times where live data is available.
- Notify drivers about charging delays.
- Display estimated costs and durations before booking.

9. Lack of Nearby Available Providers

Problem: Drivers may not find an available charging provider in their immediate area, especially during emergencies.

Solution:

- Gradually increase the search radius.
- Include verified commercial charging providers.

- Suggest alternative nearby charging facilities.
- Allow future or scheduled bookings.
- Encourage provider registration in underserved areas.

10. Electrical Safety and Equipment Quality

Problem: Improper wiring, unsuitable charging equipment or overloaded electrical connections may cause electrical hazards.

Solution:

- Verify charging facilities before approval.
- Require suitable and certified charging equipment.
- Encourage electrical inspections by qualified professionals.
- Check appropriate earthing and electrical protection.
- Follow applicable electrical safety standards and local regulations.
- Suspend facilities reported as unsafe until verified.

7. FUTURE SCOPE

Initially, RapiVolt will operate as a single application with separate Driver and Provider dashboards.

As the platform grows, we plan to introduce additional features and improvements:

1. **Separate Applications:** Develop independent applications for Drivers and Charging Providers, each with dedicated interfaces and functionalities.
2. **Live Charging Tracking:** Display charging progress, battery percentage and estimated completion time where supported by charger or vehicle integration.
3. **Smart Recommendations:** Suggest providers based on distance, price, availability and charger compatibility.
4. **Online Payments:** Integrate secure payment gateways and digital receipts.
5. **Advanced Notifications:** Send real-time booking updates and charging completion alerts.
6. **Ratings and Reviews:** Build trust through user feedback.
7. **Charging History:** Maintain records of previous bookings, charging sessions and transactions.
8. **Emergency Charging Assistance:** Connect drivers with mobile charging assistance providers where available.
9. **Smart Energy Monitoring:** Provide detailed information about electricity consumption and charging costs.
10. **Charging Time Companion:** Recommend nearby cafes, restaurants, shopping areas and other useful places while the vehicle is charging.

11. **Expansion:** Extend services to more cities and charging networks.

8. OUR VISION

Our vision is to make EV charging more accessible, convenient and reliable by creating a connected charging ecosystem.

RapiVolt aims to simplify the charging experience for EV drivers while providing homeowners and professional charging providers with new earning opportunities.

RapiVolt – Making EV Charging Accessible, Anywhere.